2 distinct values

"many" distinct values

Child	ID	Insurance
no	1	yes
yes	2	yes
yes	3	yes
yes	4	yes
no	5	no
no	6	no
no	7	no
no	8	no

generalization

intuition: to choose Child for splitting

ID3

- a higher tendency to choose an attribute containing more values

For attribute Child,

$Info(Child, T) = 0.4512$

$Gain(Child, T) = Info(T) - Info(Child, T)$

$= 1 - 0.4512$

$= 0.5488$

For attribute ID,

$Info(ID, T) = 0$

$Gain(ID, T) = Info(T) - Info(ID, T)$

$= 1 - 0$

$= 1$

We must choose attribute ID

C4.5

- try to penalize an attribute containing more values

- For attribute Child,

$$SplitInfo(Child) = 0.9544$$

$$Gain(Child, T) = \frac{0.5488}{0.9544}$$

$$= 0.5750$$

- For attribute ID,

$$SplitInfo(ID) = 3$$

$$Gain(ID, T) = \frac{1}{3} = 0.33$$

We must choose attribute Child

Ex. 5.

Q1(b)

$$Info(T) = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

$$Info(T_{yes}) = 1 - 1^2 - 0^2 = 0$$

$$Info(T_{no}) = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.32$$

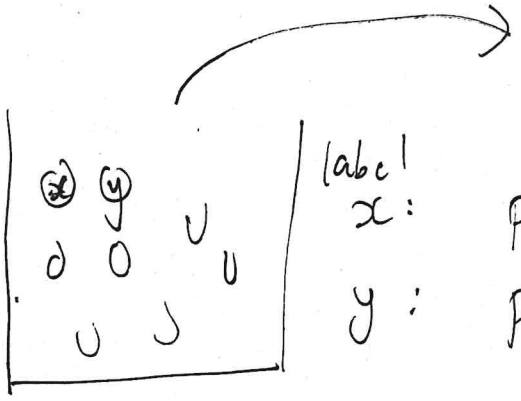
$$Info(Child, T) = \frac{3}{8} \times Info(T_{yes}) + \frac{5}{8} \times Info(T_{no})$$

$$= 0.2$$

$$Gain(Child, T) = Info(T) - Info(Child, T)$$

$$= \frac{1}{2} - 0.2$$

$$= 0.3$$



P

= P(two objects/balls have two different labels (or insurance values))

= 1 − P(two objects have the same label x)
− P(two objects have the same label y)

$= 1 - p_x^2 - p_y^2$

e.g. 1 more info: $p_x = 1.0$, $p_y = 0.0$ → $P = 1 - 1^2 - 0^2 = 0$

e.g. 2 less inf.: $p_x = 0.5$, $p_y = 0.5$ → $P = 1 - 0.5^2 - 0.5^2 = 0.5$

If P is smaller, we have more inf^n

$$P(B|A) = \frac{P(BA)}{P(A)}$$

$$= \frac{P(AB)}{P(A)}$$

$$P(AB) = P(B|A)\,P(A)$$

$$P(A|B) = \frac{P(AB)}{P(B)}$$

$$= \frac{P(B|A)P(A)}{P(B)}$$

Goal

$$P(\text{Insurance} = \text{Yes} \mid \text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no})$$

$$= \frac{P(\text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no} \mid \text{Insurance} = \text{Yes}) \times P(\text{Insurance} = \text{Yes})}{P(\text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no})}$$

$$= \frac{P(\text{Race} = \text{white} \mid \text{Ins.} = \text{Yes}) \times P(\text{Income} = \text{high} \mid \text{Ins.} = \text{Yes}) \times P(\text{Child} = \text{no} \mid \text{Ins.} = \text{Yes}) \times P(\text{Ins.} = \text{Yes})}{P(\text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no})}$$

$$= \frac{(3/4)^{1/4} \times (2/4)^{0} \times (1/4)^{1} \times (4/8)^{1/2}}{P(\text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no})}$$

$$= \frac{0.046875^{0}}{P(\text{Race} = \text{white}, \text{Income} = \text{high}, \text{Child} = \text{no})}$$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

$$P(\text{Yes} \mid \cdots) + P(\text{No} \mid \cdots) = 1$$

$$\frac{0.046875}{P(\cdots)} + \frac{0}{P(\cdots)} = 1$$

$$P(\cdots) = 0.046875$$